

# Eliza Zilliax Blodget

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## Education

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**University of Chicago**, Bachelor of Arts in Physics with Honors June 2024

Honors thesis: *Perturbative Minimax Entropy Models for Synchrony in Large Neuronal Populations*

Thesis advisor: Stephanie Palmer

Graduate coursework: Soft Condensed Matter Physics; Statistical Mechanics; String Theory I-II;

Theoretical Neuroscience: Statistics and Information Theory; Quantum Field Theory I-II-III

## Research

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**Research Analyst**, Palmer Theory Group, University of Chicago Aug 2024 – Present

Dept. of Organismal Biology

PI: Stephanie Palmer

Used topologically restricted statistical physics models to study efficient, robust encoding of scene identity in naturalistic environments, analyzing data from neurons in the salamander retina. A new project focuses on boundary dynamics in finite flocks, using hydrodynamic and soft-matter frameworks.

**Undergraduate Research Assistant**, Palmer Theory Group, UChicago Sep 2023 – Jun 2024

Dept. of Physics

PI: Stephanie Palmer

Studied the Ising model and critical phenomena on complex networks. Constructed maximum entropy models of neuronal activity constrained to tree-like networks, and used these models to analyze data from large populations of neurons in the retina and hippocampus.

## Publications

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**Topologically constrained statistical physics models capture functional aspects of a sensory code** March 2025

*Eliza M. Z. Blodget*, Christopher W. Lynn, Stephanie E. Palmer

[overleaf.com/read/bkgksktfvsrw#db422d](https://overleaf.com/read/bkgksktfvsrw#db422d) (manuscript in preparation)

## Conferences, Presentations & Workshops

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**American Physical Society Global Physics Summit**, Denver, CO March 2026 (upcoming)

*Topologically constrained statistical physics models capture functional aspects of a sensory code* (Talk)

**COSYNE**, Lisbon, Portugal March 2026 (upcoming)

*Topologically constrained models capture functional aspects of a sensory code* (Poster)

**APS Global Physics Summit**, Anaheim, CA March 2025

*Inferring minimax entropy models from neural responses to natural movies* (Talk)

**NTMB Emerging Directions Workshop**, Chicago, IL February 2025

## Teaching

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- Grader**, Biological Sciences Collegiate Division, University of Chicago Mar – Jun 2024  
Grader for BIOS 20198: Biodiversity
- Teaching Assistant**, Biological Sciences Collegiate Division, UChicago Jan – Mar 2024  
Lecture and laboratory teaching assistant for BIOS 10140: Inquiry-based Exploration of Biology

## Leadership & Outreach

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- President**, UChicago Society of Women in Physics Sep 2023 – Jun 2024  
Planned and oversaw SWiP activities, including speaker events, study breaks, and board meetings.  
Worked with the Department of Physics on outreach and inclusion initiatives for underrepresented groups in physics, both at UChicago and in the surrounding community.
- Treasurer, Webmaster, First-Year Representative**, UChicago SWiP Sep 2020 – Jun 2023